CATERPILLAR®

DEALER BEST PRACTICE SERIES

Developing a Cost Management Software Tool

Maintenance and Repair | Application | Component Life Management | Component Renewal (CRC) | **MARC Management**

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1109-5.04-1259

1.0 Introduction

Mining companies have devoted special attention to cost management and profitability of service contracts in order to improve the results of their operations. Several tools and methods have been introduced as a way of incorporating the traditional accounting vision, new perspectives and analysis of the contractual costs.

Among these methods, the activity-based costing and management stand out (ABC costing). The ABC allows one to control the activities and processes as a way of improving organizational performance and the value perceived by the customer.

The incorporation of the vision of processes and activities into the costing and profitability system gives managers an important tool for better understanding of the contract business and its outcome. The products and services offered by any contract are a direct consequence of their production processes. The understanding and analysis of these processes enable the implementation of improvement initiatives to increase business profitability.

This Best Practice aims to develop a model for cost and income evaluation and management based on ABC activities in the area of Sotreq's Mining contracts and to deploy a (software) system to determine these costs.

We are going to develop a tool for decision-making where it is possible to identify the equipment maintenance cost and the contract structure to support the operation, indicating where and when to take action. In general, the solution aims to:

1. Expand reporting of results. Currently by profit center, at customer, contract, fleet and equipment level.
2. Enable comparison of the projection (which guided the contract negotiation) with the actual contract life (actual x projected)
3. Enable comparison of the projection (made by the Contract Development area) to budgeted (made by the contract manager)
4. Develop projections, using actual results to calculate a new estimate for the contract and the fleet, allowing corrections
5. Provide spreadsheet adjustments to calculate rates for new contracts

At the end of the project, we implemented cost control providing different views:

- ✓ **Cost control by sector**: Monitoring and detailed costs control of various contract sectors, with a greater level of detail than what is provided by the accounting system
- ✓ **Process costs**: Obtainment and analysis of process cost information enabling systematic proposition of improvement initiatives, increased productivity and more efficient use of resources
- ✓ **Cost per Customer**: Detailed costs per customer and contract, pointing to activities performed on each equipment
- ✓ **Costs per Equipment**: Detailed costs by equipment, enabling the review of differentiated values according to the operational complexity for each equipment
- ✓ **Hourly costs**: Personnel costs, occupancy, equipment, etc., broken down by sector, activity, work order, customer, allowing for more efficient resource management

2.0 Best Practice Description

Concepts and Theoretical basis

Based primarily on cost classifications, COSTING SYSTEMS were developed for allocating costs to products, cost centers, activities, among others.

a. Costing System:

The process that is used to accumulate costs by products or services. A Costing System is formed by joining a **principle** and a **costing** method.

b. Principles of Costing:

Basic philosophies that govern each system. The principles of costing answers the question "***what costs will be allocated to products or services?***"

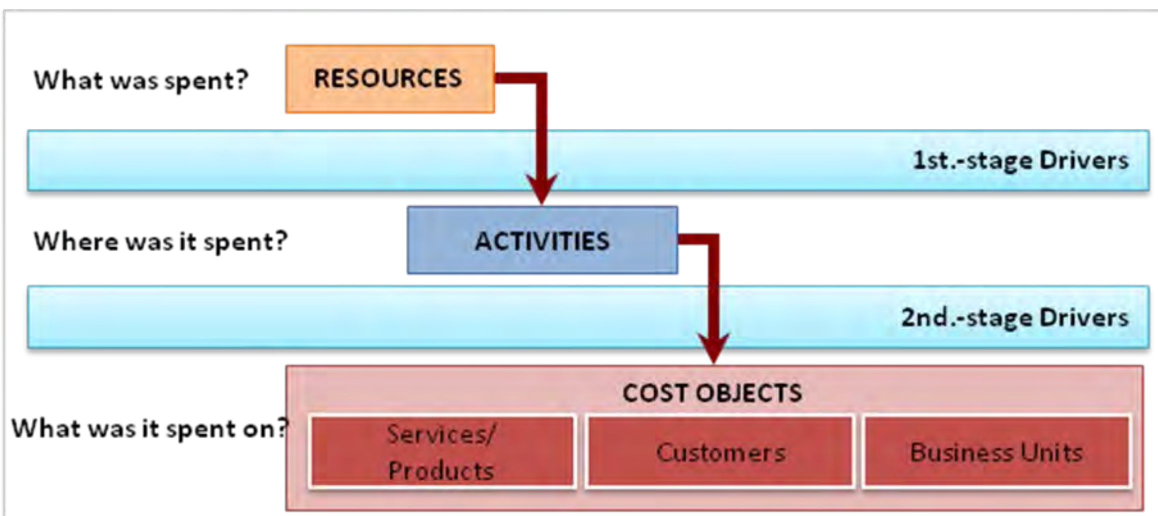
c. Costing Methods:

The mechanics of cost allocation to products or other objects. The methods answer the question "**How are costs allocated**?"

d. Method of Cost Centers:

This method splits the company into cost centers to which it allocates the costs incurred within a given period by means of predefined pro rata settings. Cost Centers are determined by factors of organization, location, responsibility and uniformity. According to the function they perform the Cost Centers can be classified as production, auxiliary, sales and common. Sotreq has its accounts organized according to this methodology.

Method of Activity-Based Costing (ABC):



ABC is a method that tracks the organization's or unit's activities and checks how these activities are related to revenue generation and resource consumption. It aims at assessing the activities carried out by a company using vectors (drivers) to allocate them in a more realistic way to products and services. This will be the methodology adopted in the modeling of this solution.

Description of Methodology

As initial project scope, we elected a MARC (*Maintenance and Repair Contract*) set with a customer of Sotreq's Mining Business Unit. After the initial project, the PBTH (*Power By The Hour*) and MARC contracts of other customers will be added. All cost centers (administrative, operational and financial were considered in the data survey and financial analysis. The modeling was developed considering all the possibilities of service contracts that exist today in Sotreq.

The original structure of cost centers and accounts of the Mining Business Unit were adopted, as well as the hierarchy of cost viewing, namely: customer, contract, fleet, equipment. At a later stage, the possibility of extending the hierarchy up to the component level will be evaluated.

Next, the resources, activities and objects were identified and set with their interrelationships.

a. Resources:

Resources are the cost elements of the activity-based costing. Each resource represents a type of cost or expense that will be driven to the activities.

Under the current accounting plan used by Sotreq, the following groups of financial accounts were considered as resources:

CCOST	BW ACCOUNTS	DRIVER
	Gross Sale	Assessment worksheet + BW -> directly to equipment
	Sales Taxes	Assessment worksheet + BW -> directly to equipment
	Sale Cost	DBS by OS -> directly to equipment (OS's as applicant)
	Expenses Inter Parts	DBS by OS -> directly to equipment (OS's as executor and applicant)
	Inter Services Expenses	DBS by OS -> directly to equipment (OS's as applicant)
	Personnel Expenses	Payroll spreadsheet per employee -> for the positions
	Other Operating Expenses	Prorate expenditure on staff (separate operational from administrative)
	Cost of Guarantee	
	Operating Income	
	Financial Expenses	

b. Activities:

The activities are combinations of human, material, financial and technological resources to produce goods or services.

The following activities were considered in the modeling:

CCOST	ACTIVITY	DRIVER
OPERATION	**Productive**	
	Field Inspection	
	PM and Backlog	
	Workshop Correction	
	Field Correction	
	Lubrication	Via SM Entry -> for equipment
	Unproductive	
	Displacement	
	Cleaning / Organization	
	Training	
	Idle	
ADMINISTRATIVE	Administrative Activities	Revenue / Earnings
		Maintenance hours (currently used)

c. Objects:

The cost objects are the end point where the costs are appropriated. They are the reason why an ABC model is developed in a company. They can be products, services, customers (markets), sales channels, subsidiaries, business units, or any other entity one wants to bear for several different purposes.

In the model, we considered the fleet and its equipment as end objects in the cost appropriation system:

CUSTOMER	
	FLEET
	MODEL
	EQUIPMENT
	COMPONENTS

To allocate resources to activities and objects, allocation criteria are set. Thus, the drivers represent the criteria by which the costs are allocated, from the accounting information, through Resources and Activities, up to Cost Objects.

In the system there are two categories of cost drivers, each of which represents the metrics for allocation at a certain stage:

❖ 1st-stage drivers: are the metrics used to allocate costs to activities. Determine the proportions to allocate resources to activities and to reallocate items of cost between activities (field inspection, PM, correction, lubrication, etc.).

❖ 2nd-stage drivers: are the metrics used to allocate costs to objects. They represent, in general, statistics which tell how the activities were demanded by the objects (fleet and equipment). We can cite an example: number of hours allocated to work orders, the amount produced by product, sales volume and number of visits per product.

Following the conceptual definition above, we began the specification, collection and validation of the financial and operational information necessary:

1) Initially, we listed all ledger accounts with accounting entries in 2010 in the cost centers of the contract;

2) For each account, we set the criteria for apportionment (drivers) by activity or object;

3) Next, all accounting entries of revenue, taxes, costs and monthly expenses in fiscal year 2010 were raised by account and cost center;

4) According to the drivers set, the accounting entries were redirected to the activities and, finally, to the objects (equipment);

5) With the above conceptual model validated, we carried out:

   ❖ parameter setting of the cost model in the Sollus ABM software of the Gradda company;

   ❖ input data and calculation of the test period (2010).

6) Together with Sotreq's IT staff, we set the specifications, led the development and validated the interface which provides data that automates the software input process

It was determined that this project requires a second phase which proposed improvements as well as next steps for the currently deployed model.

The visualization and dissemination of the monthly earnings will be performed through users' report inquiry and charts prepared. They will be divided among the Business Unit Board, Regional Managers, Contract Management and the Contract Development area.

After the end of the 1st phase of the project, the software will be delivered to a central administrator, with free user access, which will be set by the respective area managers.

3.0 Implementation Steps

This project had its origin in a joint study by Sotreq Mining Business Unit areas of Contract Development and Systems and Process Development located at the branch of Contagem - MG. Besides the mentioned areas, the IT Headquarters and Mining Operations also participated. Beginning in December 2010 and ending in June 2011, the project was developed and deployed in seven months.

We understand its replication is possible both to the company's other business units, as well as to other Caterpillar dealers, providing they have DBS and SAP systems in use.

The following were identified as potential risks and associated their strategies to counter or mitigate such risks, the following factors:

- ❖ Difficulties in team allocation:
  - ✓ Plan ahead of schedule.
- ❖ Scope not adhering to needs:
  - ✓ High involvement of Sotreq's project team and internal customers throughout the project.
- ❖ Poor collaboration in the support areas:
  - ✓ Involve support areas through the project sponsor.
- ❖ Difficulty in obtaining the data:
  - ✓ Diagnose the source information in advance and monitor the data collection.
- ❖ Difficulty in building interfaces and processes for automation of data supply:
  - ✓ Ensure its availability
  - ✓ Map data sources clearly
  - ✓ Set processes for manual data collection
  - ✓ Adapt model complexity to the information available

The deployment project for the cost management model was carried out according to the steps detailed below (see schedule in Annexes):

I. Step 1 - Project Planning and Management:
   a. Meet to set up expectations/earnings of the project;
   b. Meeting of control 1;
   c. Closing meeting.

II. Step 2 - Conceptual Definition of the cost model:
   a. Conceptual Definition of the cost model:
   b. Definition of resources, activities, objects and their correlation;
   c. Specification of the necessary financial and operational information;
   d. Raising of the necessary financial and operational information;

e. Validation of the necessary financial and operational information;

III. Step 3 - Parameter Setting and Validation of the Sollus ABM Cost Model:

a. Validation of the conceptual model;

b. Parameter setting of the Sollus ABM Cost Model:

c. Input data and calculation of one test period (2010).

IV. Step 4 - Training in Sollus ABM and Knowhow Transfer:

a. Presentation and validation of the cost model in Sollus ABM;

b. Training in Sollus ABM.

V. Step 5 - Preparation Of Permanent Interfaces and Proposal of Improvements:

a. Specification of permanent interfaces to provide data;

b. Formulation of permanent interfaces to provide data;

c. Validation of permanent interfaces to provide data;

d. Proposals for improvements and next steps for evolution of the model.

VI. Step 6 - Findings of the 5 subsequent months:

a. Findings of the subsequent months (January-May 2011);

b. Monitoring the findings of the 5 following months (January-May 2011).

4.0 Benefits

Often, business managers make important decisions about pricing, product and customer mix, allocations of resources and budgets based on information provided by the company's costing system.

Activity-Based Management (ABM) is a method that allows companies to control their activities and processes in order to improve organizational performance and value perceived by your customer.

The ABM can track the company's costs with greater accuracy than traditional methods. Some of the benefits that may result from using the ABC methodology:

- Cost finding of each individual equipment, service contract, customer, or process with a high accuracy level
- Obtainment of useful operational control information
- Routing existing contracts correction through initiatives to reduce costs, fleet change, addition of machines, contract renegotiations
- Improve the pricing model for new contracts.

The ABM model is based on the premise that the company's resources be consumed by processes and activities. Likewise, the company's processes and activities exist only to serve, directly or indirectly, the demand for their products, services and customers. The implementation of an ABC cost management model will provide multiple benefits including:

- Cost Control: Monitoring and detailed process of cost control services and contract, as a whole. Source analysis and cost-generating factors, providing feedback to reduce costs
- Profitability: Identification of equipment and fleets, contracts and more, less profitable business customers, providing information for well-founded decision-making
- Process improvement: Obtainment and analysis of information on the process costs enable systematic proposition of improvement initiatives, increased productivity and more efficient use of resources
- Feedback for planning: Using the information provided by the model of cost management, managers now have a tool to set the company's strategic and financial planning
- Pricing: Detailed analysis of unit costs for pricing of service contracts
- Analysis of outsourcing: Information for decision making about outsourcing parts of the production process
- Simulation Scenarios: Through the simulation scenarios, it enables impact evaluation of initiatives and action plans before putting them into practice;
- Dissemination of information: Possibility of Online availability of financial information and performance indicators for various Mining managers.

5.0 Resources Required

For this project, Sotreq invested in hiring a consulting and software development company:

- ✓ Consulting Hours: 160h;
- ✓ Investment: R$ 32,000.00 (thirty-two thousand reais);

Sotreq’s team was composed of 2 dedicated people 1 week per month for 5 months.

It was not necessary to acquire specific hardware to install the software, because computers used by Sotreq already have the required operational capacity. Nor was it necessary to hire additional manpower to operate and/or use the software because the assignments were absorbed in the daily work loads. The project team involved in the project participated in training which was managed and implemented by the software development company; participated in replicating the training of new users in-house.

6.0 Supporting Attachments / References

- ✓ Activity Module Screen;
- ✓ Object Module Screen;
- ✓ Hourly Operated Rate Report;
- ✓ Comparative Report of Earnings by Fleet Equipment;
- ✓ Hourly Operated Rate Report;
- ✓ Fleet Report Annualized by Operated Hour.

7.0 Related Best Practices

N/A

8.0 Acknowledgements

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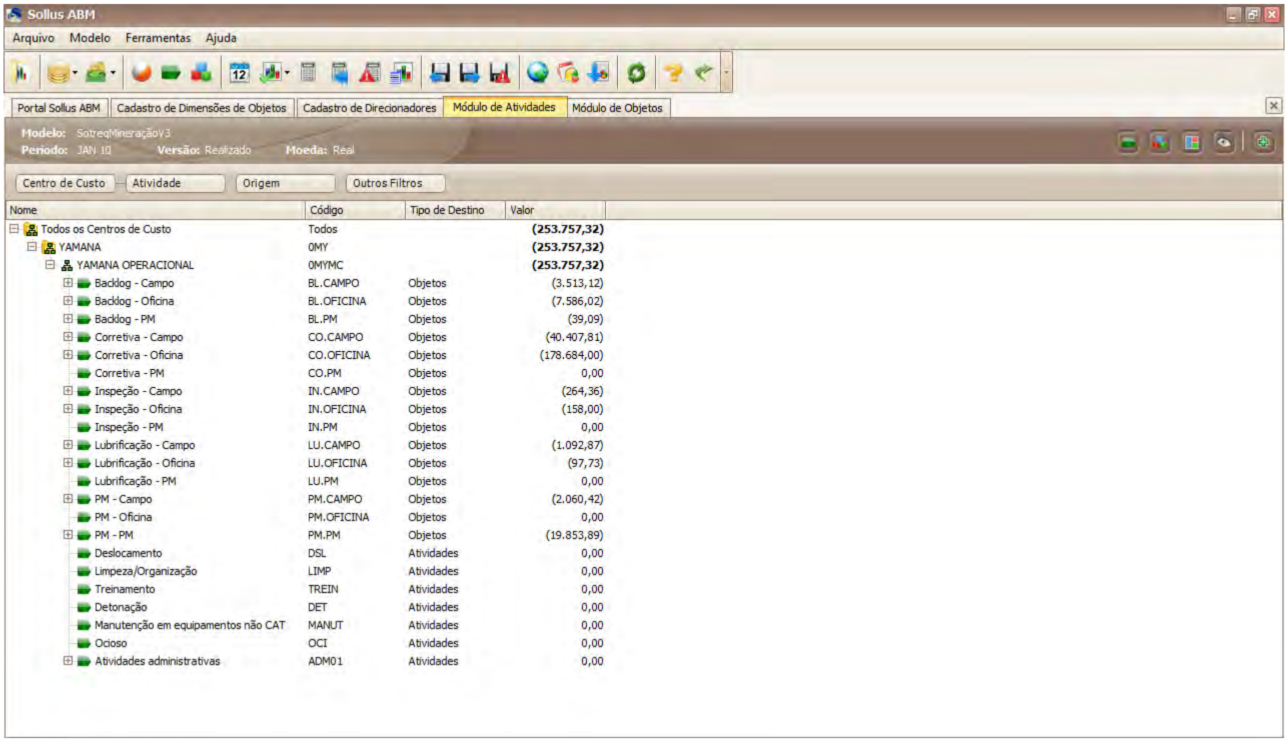
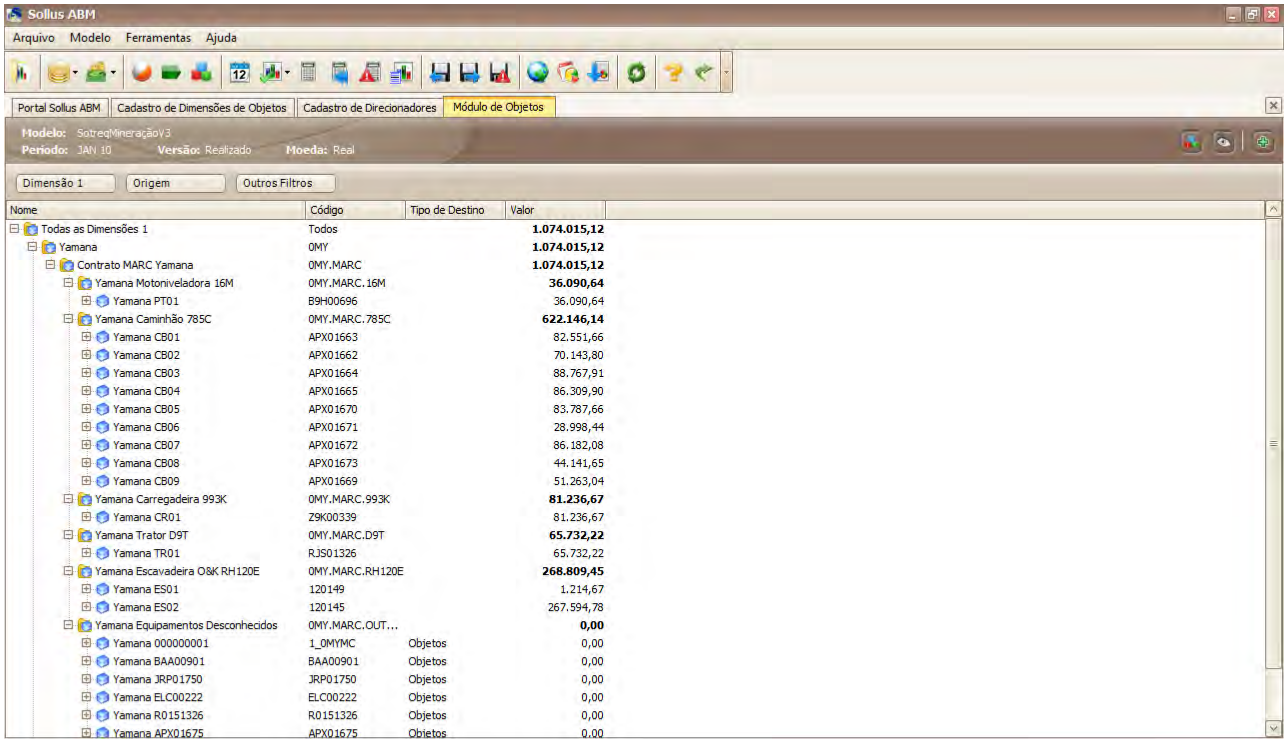
Activity Module Screen



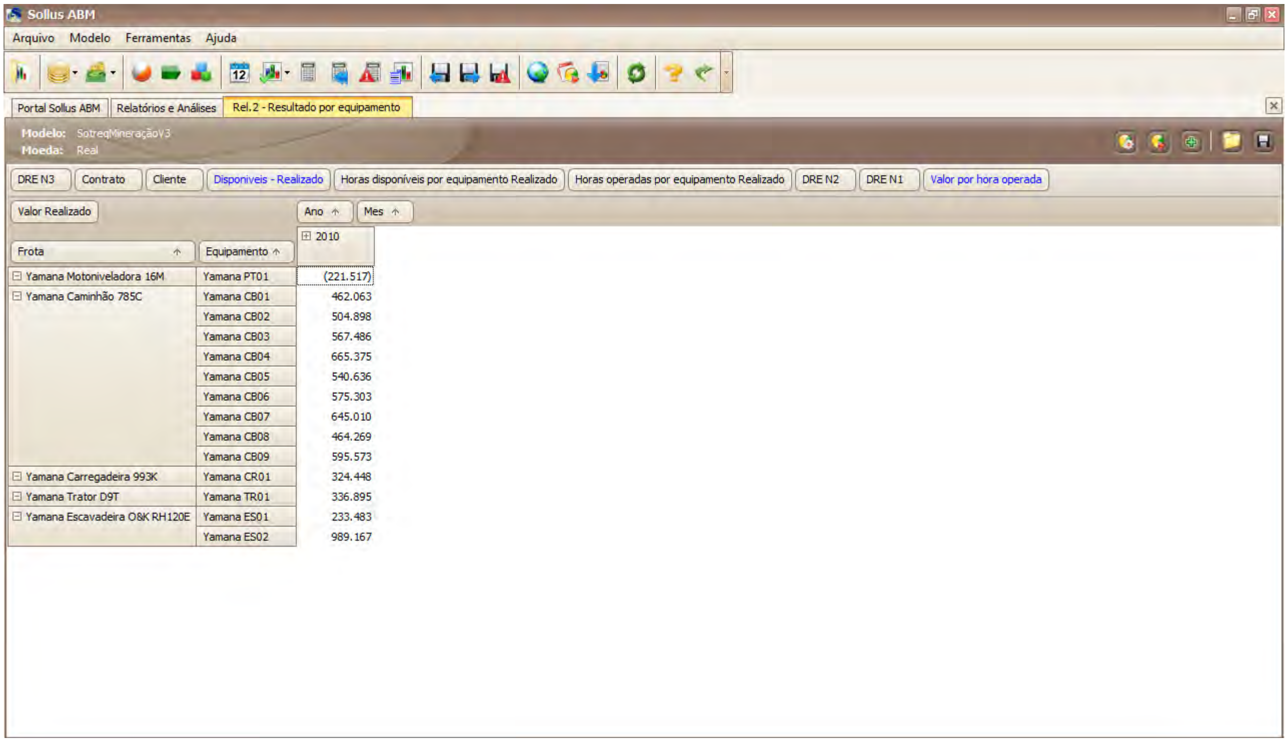
Nome	Código	Tipo de Destino	Valor
Todos os Centros de Custo	Todos		**(253.757,32)**
YAMANA	0MY		**(253.757,32)**
YAMANA OPERACIONAL	0MYMC		**(253.757,32)**
Backlog - Campo	BL.CAMPO	Objetos	(3.513,12)
Backlog - Oficina	BL.OFICINA	Objetos	(7.586,02)
Backlog - PM	BL.PM	Objetos	(39,09)
Corretiva - Campo	CO.CAMPO	Objetos	(40.407,81)
Corretiva - Oficina	CO.OFICINA	Objetos	(178.684,00)
Corretiva - PM	CO.PM	Objetos	0,00
Inspeção - Campo	IN.CAMPO	Objetos	(264,36)
Inspeção - Oficina	IN.OFICINA	Objetos	(158,00)
Inspeção - PM	IN.PM	Objetos	0,00
Lubrificação - Campo	LU.CAMPO	Objetos	(1.092,87)
Lubrificação - Oficina	LU.OFICINA	Objetos	(97,73)
Lubrificação - PM	LU.PM	Objetos	0,00
PM - Campo	PM.CAMPO	Objetos	(2.060,42)
PM - Oficina	PM.OFICINA	Objetos	0,00
PM - PM	PM.PM	Objetos	(19.853,89)
Deslocamento	DSL	Atividades	0,00
Limpeza/Organização	LIMP	Atividades	0,00
Treinamento	TREIN	Atividades	0,00
Detonação	DET	Atividades	0,00
Manutenção em equipamentos não CAT	MANUT	Atividades	0,00
Ocioso	OCI	Atividades	0,00
Atividades administrativas	ADM01	Atividades	0,00

Object Module Screen

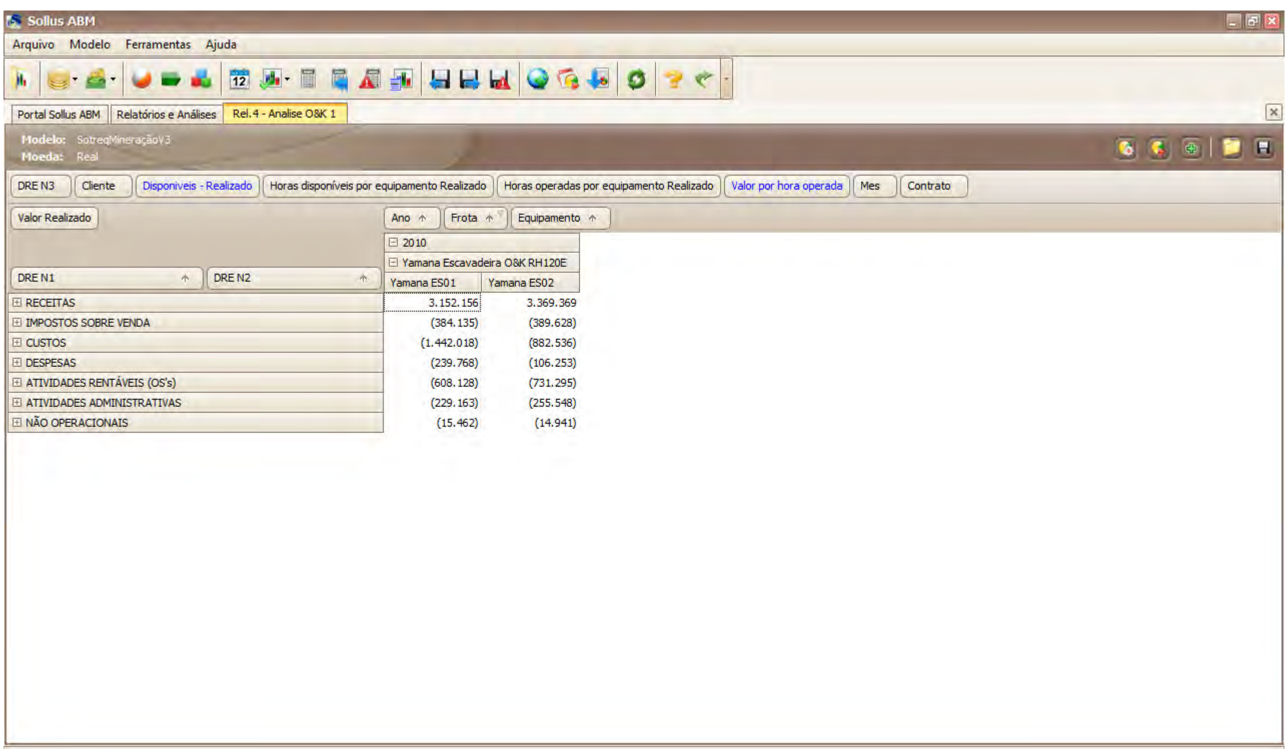


Nome	Código	Tipo de Destino	Valor
Todas as Dimensões 1	Todos		**1.074.015,12**
Yamana	0MY		**1.074.015,12**
Contrato MARC Yamana	0MY.MARC		**1.074.015,12**
Yamana Motoniveladora 16M	0MY.MARC.16M		**36.090,64**
Yamana PT01	B9H00696		36.090,64
Yamana Caminhão 785C	0MY.MARC.785C		**622.146,14**
Yamana CB01	APX01663		82.551,66
Yamana CB02	APX01662		70.143,80
Yamana CB03	APX01664		88.767,91
Yamana CB04	APX01665		86.309,90
Yamana CB05	APX01670		83.787,66
Yamana CB06	APX01671		28.998,44
Yamana CB07	APX01672		86.182,08
Yamana CB08	APX01673		44.141,65
Yamana CB09	APX01669		51.263,04
Yamana Carregadeira 993K	0MY.MARC.993K		**81.236,67**
Yamana CR01	Z9K00339		81.236,67
Yamana Trator D9T	0MY.MARC.D9T		**65.732,22**
Yamana TR01	RJS01326		65.732,22
Yamana Escavadeira O&K RH120E	0MY.MARC.RH120E		**268.809,45**
Yamana ES01	120149		1.214,67
Yamana ES02	120145		267.594,78
Yamana Equipamentos Desconhecidos	0MY.MARC.OUT...		**0,00**
Yamana 000000001	1_0MYMC	Objetos	0,00
Yamana BAA00901	BAA00901	Objetos	0,00
Yamana JRP01750	JRP01750	Objetos	0,00
Yamana ELC00222	ELC00222	Objetos	0,00
Yamana R0151326	R0151326	Objetos	0,00
Yamana APX01675	APX01675	Objetos	0,00

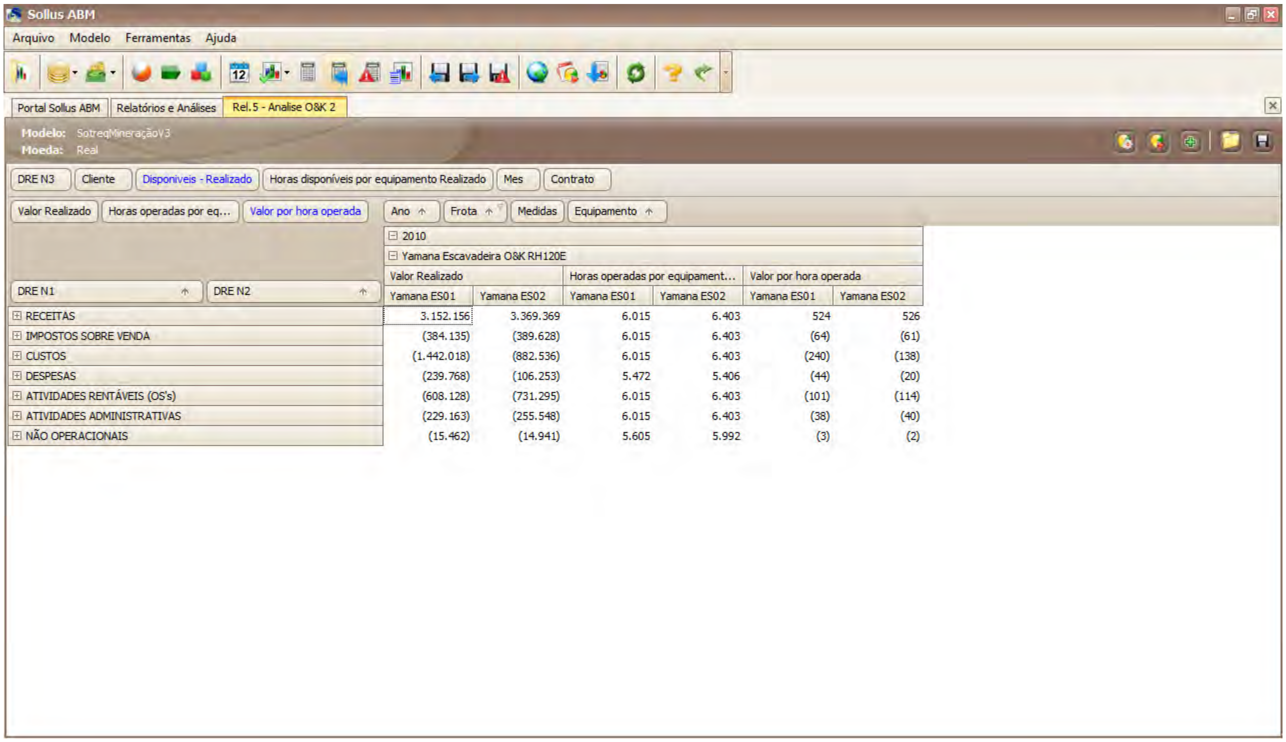
Value Report by Hourly Operated Rate



Comparative Report of Earnings by Equipment Within Fleet

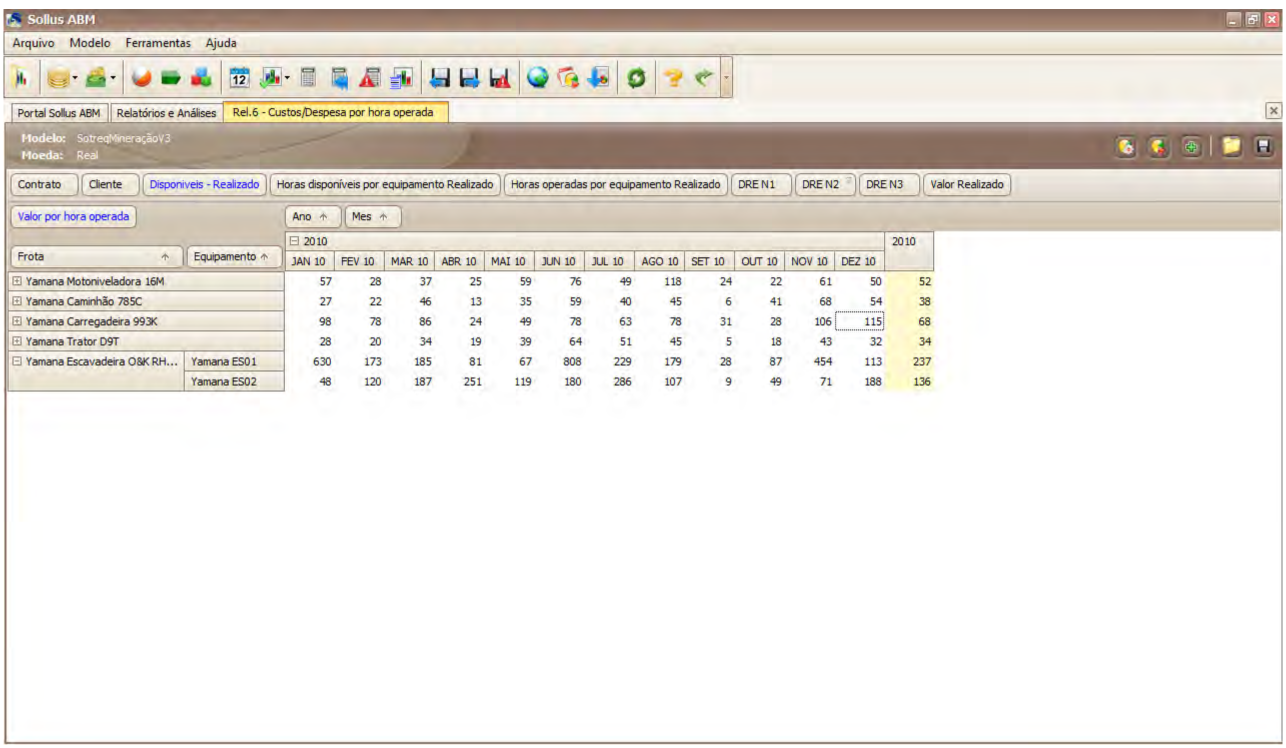


Value Report by Hourly Operated Rate



DRE N1	2010 – Yamana Escavadeira O&K RH120E: Valor Realizado – Yamana ES01	Valor Realizado – Yamana ES02	Horas operadas por equipament... – Yamana ES01	Horas operadas por equipament... – Yamana ES02	Valor por hora operada – Yamana ES01	Valor por hora operada – Yamana ES02
RECEITAS	3.152.156	3.369.369	6.015	6.403	524	526
IMPOSTOS SOBRE VENDA	(384.135)	(389.628)	6.015	6.403	(64)	(61)
CUSTOS	(1.442.018)	(882.536)	6.015	6.403	(240)	(138)
DESPESAS	(239.768)	(106.253)	5.472	5.406	(44)	(20)
ATIVIDADES RENTÁVEIS (OS's)	(608.128)	(731.295)	6.015	6.403	(101)	(114)
ATIVIDADES ADMINISTRATIVAS	(229.163)	(255.548)	6.015	6.403	(38)	(40)
NÃO OPERACIONAIS	(15.462)	(14.941)	5.605	5.992	(3)	(2)

Annualized Fleet Report by Hourly Operated Rate



Frota	Equipamento	JAN 10	FEV 10	MAR 10	ABR 10	MAI 10	JUN 10	JUL 10	AGO 10	SET 10	OUT 10	NOV 10	DEZ 10	2010
Yamana Motoniveladora 16M		57	28	37	25	59	76	49	118	24	22	61	50	52
Yamana Caminhão 785C		27	22	46	13	35	59	40	45	6	41	68	54	38
Yamana Carregadeira 993K		98	78	86	24	49	78	63	78	31	28	106	115	68
Yamana Trator D9T		28	20	34	19	39	64	51	45	5	18	43	32	34
Yamana Escavadeira O&K RH...	Yamana ES01	630	173	185	81	67	808	229	179	28	87	454	113	237
	Yamana ES02	48	120	187	251	119	180	286	107	9	49	71	188	136